

Gavin Xue

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EDUCATION

University of Waterloo

GPA: 3.9/4.0

Bachelor of Computer Science

Sep 2023 – Apr 2028

- Awarded **1 of 10** René Descartes National Scholarships (**\$20,000**) from the Faculty of Mathematics
- Relevant courses: Operating Systems, Databases, Concurrent and Parallel Programming, Artificial Intelligence

EXPERIENCE

Shopify

Toronto, ON

Software Engineering Intern

May 2026 – Aug 2026

- Shipped a policy versioning and migration framework in **Ruby on Rails** with fail-closed CI guards and idempotent **Bigtable** migration tasks, migrating 430K+ Shopify Payments records through staged production rollouts
- Improved the consistency model of a protobuf **RPC** API by adding read-your-write guarantees using primary-cluster reads, eliminating stale-read errors and 15s retry latency caused by **Bigtable** multi-cluster replication lag
- Engineered backpressure controls for bulk workloads on **Redis**-backed distributed job workers, throttling 1.4M+ record migration runs onto low-priority queues by queue depth so backfills never starved live **Kafka** signal processing

Software Engineering Intern

Sep 2025 – Dec 2025

- Unified UI components across 5 buyer identities in **React** and **TypeScript** on Shopify's checkout platform serving 2M+ merchants, advancing a P1 initiative to collapse 80+ forked components into shared base components
- Designed a unified state layer by lifting identity logic into **React** context, eliminating 7K+ lines of forked code
- Resolved a production bug that silently unenrolled buyers from Shop Pay, improving checkout conversion rates

UW Blueprint

Waterloo, ON

Software Developer

May 2025 – Present

- Led design of the platform's first file upload service, shipping a 25 MB **GraphQL** multipart upload mutation that rolls back **PostgreSQL** and **Firebase** storage writes across 3 failure paths
- Built reviewer allocation service using a round-robin algorithm in **Express.js**, removing ~2h of manual triage time
- Hardened applicant data access across 2,000+ records using **GraphQL** resolvers to implement authorization checks

Environment and Climate Change Canada

Toronto, ON

Software Developer Intern

Jan 2025 – Apr 2025

- Launched climate forecasting product for 400+ researchers, using **React** and **D3.js** for interactive data visualizations
- Built data processing pipeline with **pandas** and **Matplotlib**, automating generation of 1,000+ contour plots and structured JSON outputs for efficient client-side querying
- Refactored legacy code to follow **MVC architecture**, improving modularity and reducing the codebase by 500+ lines

Develop for Good

Remote

Software Developer

Oct 2024 – Feb 2025

- Created geospatial visualization platform with **Next.js**, showcasing 5,000+ locations with real-time, actionable insights
- Designed ETL pipeline unifying 3 data sources into **Supabase** database, using **Python** scripts and **GPT** to parse unstructured text across 50,000+ records

PROJECTS

notefuse 📄 | *Next.js, React, PostgreSQL, Redis, AWS S3, Pinecone, OpenAI*

- Built RAG-based PDF analysis chatbot using **Next.js** and **LangChain**, implementing document chunking and vector similarity search across embedded text within **Pinecone** vector database
- Reduced database queries using **Redis** caching layer to store user sessions and frequently accessed data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Ruby, C++, C, Java, SQL, HTML, CSS, Bash

Frameworks: React, Node.js, Express.js, Ruby on Rails, MongoDB, PostgreSQL, GraphQL, pandas, Matplotlib

Infrastructure & Tools: Google Cloud, Kafka, Redis, Docker, Kubernetes, Linux, Git, AWS (Certified Cloud Practitioner 🎓)